

# The Radial Lamb–Oseen Similarity Class: Uniqueness, Exact Particle Angles, and Dissipation Boundaries

Route-A source-local certificate HCS-C299

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## Abstract

We classify the bounded-at-origin, finite-circulation, radial forward-self- similar solutions of the planar viscous vorticity equation: the signed Lamb–Oseen Gaussian is the entire declared class. We reconstruct its velocity, integrate every positive-radius particle path, and give all radial moments, finite  $L^p$  norms, enstrophy, and palinstrophy.

## 1 Declared class and complete classification

On  $\mathbb{R}^2$  consider

$$\partial_t \omega + u \cdot \nabla \omega = \nu \Delta \omega, \quad \nabla \cdot u = 0, \quad \text{curl } u = \omega, \quad (1)$$

with the decaying planar Biot–Savart velocity. Fix  $\Gamma \in \mathbb{R}$ ,  $\nu > 0$ , and  $\tau = t + \tau_0 > 0$ , where  $\tau_0 \geq 0$ .

**Theorem 1** (Radial forward-similar uniqueness). *Suppose a classical radial solution of (1) has*

$$\omega(x, t) = \tau^{-1} F(\xi), \quad \xi = |x|/\sqrt{\tau}, \quad (2)$$

where  $F \in C^2([0, \infty))$  is bounded at the origin and  $\int_0^\infty |F(\xi)| \xi \, d\xi < \infty$ . If  $\int_{\mathbb{R}^2} \omega \, dx = \Gamma$ , then

$$\omega(x, t) = \frac{\Gamma}{4\pi\nu\tau} \exp\left(-\frac{|x|^2}{4\nu\tau}\right). \quad (3)$$

Thus (3) is the entire declared similarity class, including  $\Gamma = 0$ .

*Proof.* Radial vorticity has radial gradient, whereas its Biot–Savart velocity is tangential; hence  $u \cdot \nabla \omega = 0$ . Substitution of (2) into the heat equation yields

$$\nu(F'' + \xi^{-1}F') + F + \frac{\xi}{2}F' = 0. \quad (4)$$

Multiplying by  $\xi$  and integrating once gives

$$\nu\xi F'(\xi) + \frac{\xi^2}{2}F(\xi) = C. \quad (5)$$

The origin hypotheses make both left-hand terms tend to zero, so  $C = 0$ . Therefore  $F = Ae^{-\xi^2/(4\nu)}$ . The normalization  $\Gamma = 2\pi \int_0^\infty F(\xi)\xi \, d\xi = 4\pi\nu A$  fixes  $A$  and proves (3).  $\square$

The scope of Theorem 1 is essential: it classifies every profile satisfying (2) and the displayed regularity and integrability conditions, not arbitrary vortex filaments or arbitrary solutions of (1). The classical Lamb–Oseen mechanism is source owned by Oseen; we claim no historical priority.

## 2 Velocity and exact Lagrangian dynamics

Polar circulation reconstructs the velocity without an unrecorded constant:

$$u_r = 0, \quad u_\theta(r, t) = \frac{\Gamma}{2\pi r} \left(1 - e^{-r^2/(4\nu\tau)}\right) \quad (r > 0), \quad u(0, t) = 0. \quad (6)$$

Indeed  $2\pi r u_\theta = 2\pi \int_0^r s \omega(s, t) ds$ ; near zero,  $u_\theta = \Gamma r / (8\pi\nu\tau) + O(r^3)$ .

**Proposition 1** (Every particle orbit). *For  $r_0 > 0$ , radius is invariant. Put  $a = r_0^2/(4\nu)$  and*

$$\mathcal{F}_a(\tau) = \tau - \tau e^{-a/\tau} - a \operatorname{Ei}(-a/\tau). \quad (7)$$

*Between positive ages  $\tau_s < \tau_t$ ,*

$$\theta(\tau_t) - \theta(\tau_s) = \frac{\Gamma}{2\pi r_0^2} [\mathcal{F}_a(\tau_t) - \mathcal{F}_a(\tau_s)]. \quad (8)$$

*Here  $\theta$  denotes a continuous real-valued lift of polar angle, not only its class modulo  $2\pi$ . The origin is a fixed particle. If  $\Gamma \neq 0$  and  $r_0 > 0$ , then*

$$\theta(t) = \frac{\Gamma}{8\pi\nu} \log \tau + O(1) \quad (\tau \rightarrow \infty). \quad (9)$$

*Proof.* Equation (6) gives  $\dot{r} = 0$  and

$$\dot{\theta} = \frac{\Gamma}{2\pi r_0^2} (1 - e^{-a/\tau}).$$

Direct differentiation shows  $\mathcal{F}'_a = 1 - e^{-a/\tau}$ , proving (8). The continuous origin value proves the fixed-point claim. Finally  $1 - e^{-a/\tau} = a/\tau + O(\tau^{-2})$ , which gives (9) because  $a/r_0^2 = 1/(4\nu)$ .  $\square$

## 3 All moments, norms, and exact dissipation

The change of variable  $y = r^2/(4\nu\tau)$  evaluates the whole parameter family, rather than a finite selection.

**Proposition 2.** *For every integer  $k \geq 0$  and every real  $1 \leq p < \infty$ ,*

$$\int_{\mathbb{R}^2} |x|^{2k} \omega(x, t) dx = \Gamma k! (4\nu\tau)^k, \quad (10)$$

$$\|\omega(\cdot, t)\|_p^p = \frac{|\Gamma|^p}{p(4\pi\nu\tau)^{p-1}}. \quad (11)$$

*Also  $\|\omega\|_\infty = |\Gamma|/(4\pi\nu\tau)$  and*

$$\int_{\mathbb{R}^2} \omega^2 dx = \frac{\Gamma^2}{8\pi\nu\tau}, \quad (12)$$

$$\int_{\mathbb{R}^2} |\nabla \omega|^2 dx = \frac{\Gamma^2}{16\pi\nu^2\tau^2}, \quad (13)$$

$$\frac{d}{dt} \int_{\mathbb{R}^2} \omega^2 dx = -2\nu \int_{\mathbb{R}^2} |\nabla \omega|^2 dx. \quad (14)$$

*Proof.* Polar integration and  $y = r^2/(4\nu\tau)$  reduce (10) to  $\int_0^\infty y^k e^{-y} dy = k!$  and (11) to  $\int_0^\infty e^{-py} dy = 1/p$ . The supremum occurs at the origin. The case  $p = 2$  gives (12); differentiating (3) radially and integrating gives (13). Differentiation in  $t$  closes (14).  $\square$

## Source lineage

Oseen's 1912 line-vortex work is the classical owner token for the profile. Galloway and Wayne provide modern mathematical context for the two-dimensional Oseen vortex and its stability. These citations establish lineage, not a novelty or priority claim for the classical formula.

## References

- [1] C. W. Oseen, “Über Wirbelbewegung in einer reibenden Flüssigkeit,” *Arkiv för Matematik, Astronomi och Fysik* 7 (1912), no. 14.
- [2] T. Gallay and C. E. Wayne, “Global stability of vortex solutions of the two-dimensional Navier–Stokes equation,” *Communications in Mathematical Physics* 255 (2005), 97–129. DOI: [10.1007/s00220-004-1254-9](https://doi.org/10.1007/s00220-004-1254-9).